

Diazepam

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Introduction

The second **benzodiazepine** to be marketed, diazepam, was approved for human use in the early 1960s. Because of its relative safety and selectivity as an anxiolytic, it was readily accepted by the medical community, displacing the barbiturates and other sedative-hypnotics as the drug of choice for treating anxiety. For many years diazepam was one of the most widely prescribed drugs in the United States [Hobbs et al \(1995\)](#). It has sedative, hypnotic, muscle relaxant and anticonvulsant properties such that it is also used in the treatment of status epilepticus.

Diazepam, like other members of the benzodiazepine family of drugs, has a high affinity for the benzodiazepine binding site that forms part of the **gamma-aminobutyric acid-A** (GABA_A) receptor complex. It therefore potentiates the actions of GABA by enhancing inhibitory synaptic transmission. Indeed, diazepam has played a major role in defining the site of action and the physiological and behavioral responses to the benzodiazepines. Radiolabeled diazepam has been widely employed for characterizing both the central (GABA_A/benzodiazepine) and peripheral benzodiazepine binding sites. It has also been used as the labeling ligand for high throughput screens. Given the amount of research conducted with diazepam and its extensive use as a therapeutic agent, it is routinely employed as a standard in research programs aimed at testing new chemical entities as possible anxiolytics and anticonvulsants.

Nomenclature

Name of the Clinical Form

Related Names

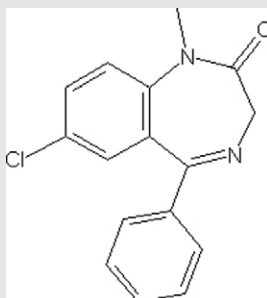
Source: [EMTREE](#)

Diazepam; Valium (trade); alupram; ansiolin; apaurin; apaurine; apozeepam; assival; audium; 1,4 benzodiazepin 2one,7 chloro 1,3 dihydro 1 methyl 5 phenyl 2h; bialzepam; bialzegan; calmpose; cercin; cercine; cersine; chlordiazeepam; 7 chloro 1,3 dihydro 1 methyl 5phenyl (2h) 1,4 benzodiazepin 2 one; 7 chloro 1,3 dihydro 1 methyl 5 phenyl(2h)1,4 benzodiazepin 2 one; 7 chloro 1,3 dihydro 1 methyl 5 phenyl 2h 1,4benzodiazepin 2 one; diastat; diazelium; diazemuls; diazidem; duxen; eridan; eurosan; evacalm; fanstan; faustan; gewacalm; la 111; la iii; lamra; lembrol; lipodiazeepam; lorinon; methyldiazepinon; methyldiazepinone; morosan; neocalme; neurolytril; noan; novazam; nsc 77518; paceum; plidan; psychopax; relanium; ro 05 2807; ro 5 2807; sedapam; seduxen; serendin; setonil; sonacon; stesolid; stesolid novum; stesolin; tanquo tablinen; tranimul; tranquo puren; umbrium; valaxona; valiquid; valium; valium mixed micelles; valium mm; valrelease; vatran; vival; vivol; wy 3467;

Chemical Names	zetrax; 1,4 benzodiazepin 2 one,7 chloro 1,3 dihydro 1 methyl 5 phenyl 2h; 7 chloro 1,3 dihydro 1 methyl 5 phenyl (2h) 1,4 benzodiazepin 2 one; 7 chloro 1,3 dihydro 1 methyl 5 phenyl (2h)1,4 benzodiazepin 2 one; 7 chloro 1,3 dihydro 1 methyl 5 phenyl 2h 1,4 benzodiazepin 2 one; antenex; ducene; valpam
CAS Number	439-14-5

Basic Chemistry

Chemical Structure



Chemical Formula

C₁₆ H₁₃ Cl N₂ O

Properties

Physical Properties

Diazepam is a colorless crystalline compound with a slight bitter taste and is insoluble in water.

Molecular Weight

284.745

Solubility

While diazepam is insoluble in water. It is soluble in chloroform, DMF, benzene, acetone and ethanol.

Ionization Constant

	Value	Salt	Conditions	Reference	Comments
pKa	3.4	Base		The Merck Index (1996)	

Human Pharmacokinetics

Diazepam is absorbed rapidly and completely after oral administration and rapidly penetrates into brain. Peak plasma concentrations are achieved at 15–90 minutes for adults and 15–30 minutes for children, with a secondary peak 6–12 hours after ingestion due to first-pass metabolism. It is extensively metabolized in the **liver**, producing three active products, such as nordiazepam, which are pharmacologically active with a plasma half life of 30–200 hours. Nordiazepam itself is converted to the active metabolite **oxazepam**, which has a mean plasma half life of 9 hours. **Temazepam** is the third active

metabolite formed, with a plasma half life of 10–20 hours. The formation of active metabolites is responsible, in part, for the long duration of action of diazepam. Diazepam does not appear to induce microsomal enzyme activity so the rate of its metabolism, and of other sedative-hypnotics, is not appreciably changed following chronic administration of this agent. Diazepam is widely and rapidly distributed throughout the body and is highly bound to albumin in the plasma. The metabolites are eliminated by conjugation with **glucuronic acid**, and are excreted in urine.

Pharmacokinetic Properties

	Value	Units	Prep. and Route of Admin.	Reference	Comments
Absorption					
Bioavailability	100	%	p.o.	Hobbs et al (1995)	
Distribution					
Volume of Distribution	1.1	l/kg		Hobbs et al (1995)	
Plasma Protein Binding	99	%		Hobbs et al (1995)	
Metabolism					
Plasma Half-Life	43	hrs		Hobbs et al (1995)	
Bio Half-Life					
Clearance	0.38	ml/min/kg		Hobbs et al (1995)	
Routes of Elimination	Excreted in the urine.				

Targets-Pharmacodynamics

Diazepam binds with high affinity to the benzodiazepine binding site component of the **GABA_A receptor**. By its attachment to this site, diazepam enhances GABA_A receptor responses to the **neurotransmitter**. The diazepam binding site is distinct from the GABA binding site on the GABA_A receptor. Inasmuch as GABA is an inhibitory neurotransmitter, diazepam facilitates GABA-mediated neuronal hyperpolarization, decreasing **central nervous system** activity. Because diazepam attaches to the benzodiazepine component of the GABA_A receptor, its pharmacological effects can be reversed by **flumazenil**, a benzodiazepine antagonist.

Target Name(s):

GABA-A receptor

Therapeutics

Diazepam is a central nervous system depressant and is useful in the management of **anxiety**. It is used clinically to treat anxiety associated with major psychiatric disorders, such as affective illness and psychosis. At higher doses diazepam is used clinically as a hypnotic, and when administered i.v., for the treatment of status epilepticus and other convulsive states such as tetanus and eclampsia. Diazepam is also used as premedication before surgical or other stressful therapeutic procedures are performed. It may also be utilized in the treatment of withdrawal symptoms to other central nervous system depressants such as **ethanol** and barbiturates.

Diazepam

Indications

	Value	Units	Prep. and Route of Admin.	Reference	Comments
Management of Anxiety (Adults)					
Dosage	2–10	mg	p.o.	Physician's Desk Reference (2003)	2–4 times/day for mild/moderate anxiety.
Muscle Spasm					
Dosage	2–10	mg	p.o.	Physician's Desk Reference (2003)	3–4 times/day
Acute Ethanol Withdrawal					
Dosage	10	mg	p.o.	Dollery (1999)	3–4 times/day. Reduce to 5 mg after 24 hrs.
Basal Sedation					
Dosage	10–30	mg	i.v.	Dollery (1999)	Before surgery or other stressful procedures
Status epilepticus and other convulsive states					
Dosage	0.15–0.25	mg/kgbody weight	i.v.	Dollery (1999)	Repeat after 10–15 min if required. Or use i.v. drip of <3 mg/kgbody weight in 24 hrs.

Contraindications

Diazepam is contraindicated in those with hypersensitivity to benzodiazepines, in those with myasthenia gravis, and those aged <6 years.

Adverse Effects

Common side effects encountered with diazepam are a result of its central nervous system depressant properties. These include sedation, ataxia and confusion. At sufficient doses diazepam can also cause anterograde amnesia. In some individuals, particularly children, diazepam may stimulate central nervous system activity, inducing hyperactivity. Continued administration of diazepam can lead to tolerance and addiction. The central nervous system depressant side effects are more insidious and, most often observed in the elderly.

Agent-Agent Interactions

Agent Name	Mode of Interaction
Barbiturates	Concomitant use of diazepam with other central nervous system depressants should be avoided since there will be at least an additive affect in decreasing central nervous system function, with an increased risk for respiratory depression. Likewise, care must be taken when administering diazepam with a narcotic analgesic since both agents depress respiratory function.
Ethanol	Concomitant use of diazepam with other central nervous system depressants should be avoided since there will be at least an additive affect in decreasing central nervous system function, with an increased risk for respiratory depression. Likewise, care must be taken when administering diazepam with a narcotic analgesic since both agents depress respiratory function.
Phenothiazines	Concomitant use of diazepam with other central nervous system depressants should be avoided since there will be at least an additive affect in decreasing central nervous system function, with an increased risk for respiratory

	depression. Likewise, care must be taken when administering diazepam with a narcotic analgesic since both agents depress respiratory function.
MAO inhibitors	Concomitant use of diazepam with other central nervous system depressants should be avoided since there will be at least an additive affect in decreasing central nervous system function, with an increased risk for respiratory depression. Likewise, care must be taken when administering diazepam with a narcotic analgesic since both agents depress respiratory function.
Phenytoin	Metabolic elimination of phenytoin may be altered by diazepam
Cimetidine	Cimetidine may delay diazepam clearance
Levodopa	Levodopa patients may see deterioration of parkinsonism

Pre-Clinical Research

Diazepam is used routinely as a standard for preclinical screens of new chemical entities such as anxiolytics, anticonvulsants and muscle relaxants. In addition, radiolabeled diazepam is utilized for studying benzodiazepine receptor binding.

Pharmacokinetics

Potency

	Value	Units	Organ/ Tissue	Prep. and Route of Admin.	Cell Line/ Type	Effects	Exp. End Point	Reference	Comments
Mouse									
LD50	970	mg/kg		p.o.					
ED50	1.5	mg/kg		i.m.		anti-convulsant		Wesseling et al (1971)	
Rat									
ED50	5–10	mg/kg		i.m.		anti-convulsant		Rossoff (1974)	
Rabbit									
LD50	8.8	mg/kg		i.v.				Diamantis and Kletzkina (1966)	
Cat									
ED50	0.5	mg/kg		i.m.		anti-convulsant		Okamoto et al (1983)	

Other Research

The affinity of [³H]diazepam for its receptor in rat brain is 5–18 nM [Benke and Mohler \(1998\)](#). It has negligible affinity for other receptors or enzymes. Recent work suggests that only those GABA_A receptors composed of alpha 1, alpha 2, alpha 3 or alpha 5 subunits, in combination with a beta and a gamma 2 are consistently responsive to benzodiazepines. The sedative and amnesic effects of diazepam appear to be mediated primarily by GABA_A receptors possessing an alpha 1 subunit, whereas the anxiolytic response requires the

presence of an alpha 2 subunit Rudolph et al (1999), Huang et al (2000), Low et al (2000). Such information could be useful in designing safer and even more selective anxiolytics that act through the benzodiazepine receptor.

<http://kidb.cwru.edu/pdsp.php>) lists 31 Ki values for diazepam at various receptors.

Diazepam can be obtained commercially from Sigma Chemical Co. (<http://www.sigma.com/>) (Cat#D9900).

Radiolabeled diazepam can be purchased from NEN Life Science (<http://lifesciences.perkinelmer.com/>) (NET 564).

Other Information – Web Sites

The PDSP Ki database: <http://kidb.cwru.edu/pdsp.php>

Sigma Chemical Co.: <http://www.sigma.com/>

NEN Life Science: <http://lifesciences.perkinelmer.com/>

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